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A. Related Works About Literature Retrieval Tools

Literature retrieval is a fundamental process in scientific research, and several well-established scientific literature search engines are widely used. Among them, Google Scholar and Semantic Scholar are two of the most popular platforms, offering literature retrieval services across all research fields. However, these platforms do not provide access to the full text of the indexed papers. In contrast, arXiv maintains its own comprehensive literature database, which not only facilitates literature retrieval but also provides full-text access. ArXiv primarily focuses on fields such as mathematics, physics, and computer science. Similarly, PubMed and PMC (PubMed Central) cater specifically to the field of medicine, offering specialized literature retrieval services. Despite their usefulness, these databases predominantly rely on keyword matching for retrieval, which often leads to imprecise results. Consequently, users are left with the burden of manually filtering out irrelevant literature to find the information they need.

B. SciPIP Proposed Ideas.

The detailed idea of demonstration shown in Table 3 is given as follows. **All the following contents in Appendix B are generated by SciPIP.** Other demonstrations of SciPIP proposed ideas are provided in the supplementary materials.

B.1. Detailed Idea

To improve Retrieval-Augmented Generation (RAG) systems’ handling of lengthy documents, the proposed “Multi-Stage Contextual Thinning with Semantic Twin Projections” algorithm systematically reduces document content through a novel dual-phase process that enhances semantic integrity and contextual relevance.

B.1.1. OVERVIEW OF THE MULTI-STAGE CONTEXTUAL THINNING PROCESS

The process consists of two main phases: Dynamic Attention-Based Document Thinning and Semantic Twin Projections. Each phase progressively refines the document, enhancing relevance and coherence before it is processed by the generative module.

B.1.2. PHASE 1: DYNAMIC ATTENTION-BASED DOCUMENT THINNING

In this initial phase, the algorithm applies attention mechanisms to focus on contextually important parts of a document:

Dual Embedding Representation

- Utilize dual embedding models where E_{global} captures overarching document themes, while E_{local} focuses on granular details.
- Compute global attention scores A_{global} and local attention scores A_{local} using the formula:

$$A_{global} = \text{softmax}(Q_{global} \cdot K_{global}^T), \quad A_{local} = \text{softmax}(Q_{local} \cdot K_{local}^T)$$

where Q, K are query and key matrices derived from E_{global} and E_{local} embeddings respectively.

Broad Stroke Thinning

- Combine these attention scores to prioritize sections of the document that maintain a balance between thematic relevance and detailed context:

$$A_{combined} = \alpha A_{global} + (1 - \alpha) A_{local}$$

where $\alpha \in [0, 1]$ is a tunable parameter to balance global and local attentions.

- Select document segments with the highest $A_{combined}$ scores to create a reduced version of the document, filtering out less relevant material while retaining the core content.

B.1.3. PHASE 2: SEMANTIC TWIN PROJECTIONS

In the second phase, the methodology refines the thinned document using semantic twin networks to further emphasize key information.